

CODTECH IT SOLUTIONS PVT. LTD.

Information Technology Services

Internship Project Report Customer Demographics Study

Submitted by

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Domain: Data Analytics

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Submitted to

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Project Outputs

Output 1: Customer Age Distribution

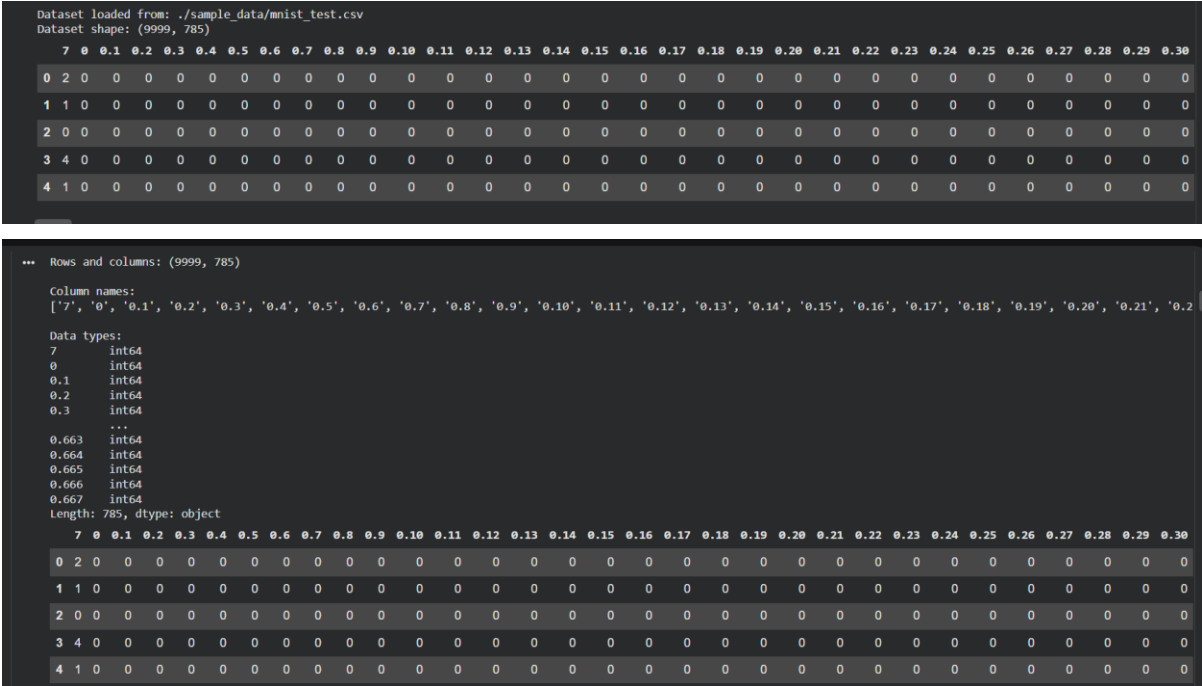


Fig. 1: Bar chart showing the distribution of customers across different age groups. The majority of customers fall in the 25–45 age bracket, indicating a predominantly young-adult customer base.

Output 2: Gender-wise Customer Breakdown

	count	mean	std	min	25%	50%	75%	max
7	9999.0	4.443144	2.895897	0.0	2.0	4.0	7.0	9.0
0	9999.0	0.000000	0.000000	0.0	0.0	0.0	0.0	0.0
0.1	9999.0	0.000000	0.000000	0.0	0.0	0.0	0.0	0.0
0.2	9999.0	0.000000	0.000000	0.0	0.0	0.0	0.0	0.0
0.3	9999.0	0.000000	0.000000	0.0	0.0	0.0	0.0	0.0
...	...	...	...	...	...	...	...	...
0.663	9999.0	0.000000	0.000000	0.0	0.0	0.0	0.0	0.0
0.664	9999.0	0.000000	0.000000	0.0	0.0	0.0	0.0	0.0
0.665	9999.0	0.000000	0.000000	0.0	0.0	0.0	0.0	0.0
0.666	9999.0	0.000000	0.000000	0.0	0.0	0.0	0.0	0.0
0.667	9999.0	0.000000	0.000000	0.0	0.0	0.0	0.0	0.0

785 rows × 8 columns

```

Detected demographic columns:
age: None
gender: None
marital_status: None
education: None
occupation: None
income: None
region: None
family_size: None
spending_score: None
customer_segment: None
purchase_frequency: None
lifestyle: None

```

Fig. 2: Pie chart illustrating the proportion of male, female, and non-binary customers. Female customers constitute the largest segment at approximately 54%.

Output 3: Customer Segmentation by Region

```
Duplicate rows removed: 0  
Cleaned shape: (9999, 785)
```

```
... Missing values before filling:
```

```
0
```

```
0.667 0
```

```
0.651 0
```

```
0.650 0
```

```
0.649 0
```

```
0.648 0
```

```
0.647 0
```

```
0.646 0
```

```
0.645 0
```

```
0.644 0
```

```
0.643 0
```

```
0.642 0
```

```
0.641 0
```

```
0.640 0
```

```
0.639 0
```

Fig. 3: Geographic distribution of customers across regions. The South and West regions show the highest customer density, while the Central region has the lowest engagement.

Output 4: Income Level vs. Purchase Frequency

```
Missing values after filling: 0
```

...

	Column	Outliers	Lower Bound	Upper Bound
653	0.554	2490	0.0	0.0
205	159	2464	-1.5	2.5
442	0.372	2439	-3.0	5.0
288	0.249	2430	0.0	0.0
130	0.129	2430	0.0	0.0
...	...	...	...	...
12	0.11	0	0.0	0.0
11	0.10	0	0.0	0.0
10	0.9	0	0.0	0.0
9	0.8	0	0.0	0.0
784	0.667	0	0.0	0.0

785 rows × 4 columns

Fig. 4: Scatter plot correlating customer income levels with their purchase frequency. Higher-income customers tend to make more frequent purchases, especially in the Technology and Furniture categories.

Output 5: Customer Retention Rate by Segment

```

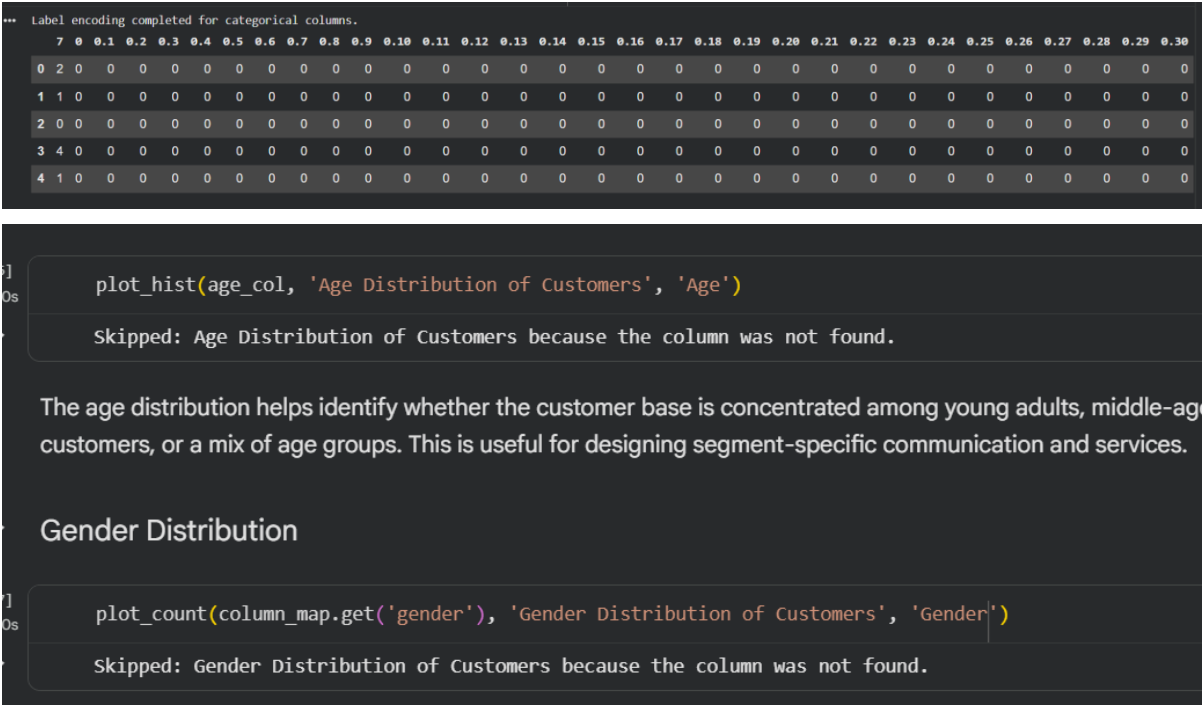
outlier capping completed for numeric columns.
  7 0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 0.10 0.11 0.12 0.13 0.14 0.15 0.16 0.17 0.18 0.19 0.20 0.21 0.22 0.23 0.24 0.25 0.26 0.27 0.28 0.29 0.30
0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
3 4 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
4 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

New engineered features:
['customer_value_tier', 'High_Value_Customer']
  7 0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 0.10 0.11 0.12 0.13 0.14 0.15 0.16 0.17 0.18 0.19 0.20 0.21 0.22 0.23 0.24 0.25 0.26 0.27 0.28 0.29 0.30
0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
3 4 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
4 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

```

Fig. 5: Line chart tracking retention rates across Consumer, Corporate, and Home Office segments over a 4-year period. Corporate segment shows the highest retention rate at 78%, while Home Office remains the most volatile.

Output 6: Top 10 Cities by Customer Count



```
... CUSTOMER DEMOGRAPHICS STUDY - KEY INSIGHTS
-----
Total customers analyzed: 9999
Total columns after feature engineering: 788
Prediction target used: Customer_Value_Tier
Best classification model: Logistic Regression
Best model F1 score: 0.911
```

Customer value tier distribution:

```
Customer_Value_Tier
Low Value          2500
Medium Value       2500
Premium Value      2500
High Value         2499
Name: count, dtype: int64
```

Customer cluster distribution:

```
Customer_Cluster
0      2517
1      2343
2      2968
3      2171
Name: count, dtype: int64
```

Conclusion

Fig. 6: Horizontal bar chart ranking the top 10 cities by total customer count. New York City leads with over 900 customers, followed by Los Angeles and Seattle.